

# Microsoft Essential PYQ Practice Pack

Product Based / Microsoft

## Study Hub Premium Placement Pack

This is original study material for preparation. It is not a copied official or proprietary company paper.

## Premium Study System

- Track: Microsoft.
- Pack type: PYQ / PYQ Practice.
- Target outcome: selection-ready confidence for SDE / Software Engineer.
- Core preparation areas: DSA, CS fundamentals, OOP/LLD, HLD basics, behavioral stories, resume projects.
- Use this pack like a mini-book: concept, table, example, practice, analysis, revision.
- Every topic should produce proof: solved questions, mock score, mistake note, interview answer, or resume evidence.

## Output Quality Rules

- Read the concept in your own words, then solve without looking at notes.
- Convert every important idea into a comparison table, checklist, or one-line rule.
- Mark important traps as Common Mistake before the mock, not after losing marks.
- Keep a PYQ-style log: question pattern, topic, time taken, mistake reason, retest date.
- Use simple English; add Hindi/Hinglish meaning in brackets when it helps recall.

## Priority Matrix

| Area | Premium Standard | Proof Before Moving On |  
| DSA | Pattern, invariant, dry run, complexity | 2 solved problems per pattern without hints |  
| CS Core | OS, DBMS, CN, OOP with examples | Explain each topic in 60 seconds |  
| Project | Problem, architecture, your contribution, tradeoff | 2-minute story plus deep-dive answers |  
| Communication | Crisp intro, structured HR stories, calm tone | 5 recorded answers reviewed once |

## Exam Tip

- Interviewers reward clean reasoning more than memorized code. Say brute force, improve it, dry run, then discuss complexity.

## Common Mistake

- Jumping into code without constraints creates wrong edge cases. Ask size, duplicates, sorted/unordered, and expected output first.

## Original PYQ Style Set

- This is original practice inspired by common placement patterns, not a copied company paper.

## Original OA and Interview Practice

- This set is designed for Microsoft style product preparation: DSA depth, CS fundamentals, design thinking, project ownership, and behavioral clarity.

## Section A: DSA Pattern Practice

- Q1. Return indices of two numbers adding to target. Explain hash map time and space complexity.
- Q2. Merge overlapping intervals and list the sorting invariant.
- Q3. Detect cycle in a linked list, then find the cycle start.
- Q4. Find shortest path in an unweighted graph and explain why BFS works.

- Q5. Find the longest substring without repeating characters. Explain sliding window movement.
- Q6. Search an element in a rotated sorted array. Explain the binary-search decision rule.
- Q7. Given daily temperatures, return days to wait for warmer temperature. Explain stack invariant.
- Q8. Count islands in a grid. Explain visited marking and boundary checks.

## Section B: DP and Optimization

- Q9. Count ways to climb stairs with 1 or 2 steps, then optimize memory.
- Q10. Find maximum subarray sum and explain why local optimum works.
- Q11. Find coin change minimum coins. Define state, transition, base case, and impossible state.
- Q12. Given jobs with start, end, profit, select maximum profit non-overlapping jobs. Explain sorting + DP idea.

## Section C: LLD and HLD

- Q13. LLD: Design a parking lot with classes, responsibilities, pricing, and edge cases.
- Q14. LLD: Design a rate limiter. Compare fixed window, sliding window, and token bucket.
- Q15. HLD: Design a tiny URL service. Cover APIs, storage, cache, collisions, and scaling.
- Q16. HLD: Design a notification system. Cover queues, retries, idempotency, and failure handling.

## Section D: CS Core and Project Depth

- Q17. Explain database indexing and when an index can hurt performance.
- Q18. Explain process vs thread and how context switching affects performance.
- Q19. Explain HTTP status codes, cookies, JWT, and session tradeoffs.
- Q20. Describe a time you handled ambiguity while preparing for Microsoft. Use a real example.

## Model Hints

- Q1 hint: hash map stores complement or seen values; time  $O(n)$ , space  $O(n)$ .
- Q3 hint: Floyd cycle detection gives meeting point; reset one pointer to head to find cycle start.
- Q8 hint: DFS/BFS from every unvisited land cell; each cell is visited once.
- Q14 hint: mention data model, request key, time bucket, concurrency, and cleanup.

## Solution Checklist

- State brute force first, then optimized approach.
- Confirm input constraints before coding.
- Dry run one normal case and one edge case.
- End with complexity and production tradeoffs.
- For design rounds, define requirements before drawing components.

## How To Review

- Mark every question as solved, guessed, skipped, or wrong.
- Rewrite the idea behind every wrong question in one sentence.
- Re-attempt wrong questions after 24 hours and again after 7 days.

## PYQ-Style Attempt Protocol

- Step 1: identify topic and difficulty before solving.
- Step 2: write the first approach, even if it is brute force or long method.
- Step 3: optimize only after the basic idea is correct.
- Step 4: dry run one normal case and one edge case.
- Step 5: add answer, time, mistake type, and retest date to the log.

## Pattern References

- Common OA patterns: hash map lookup, interval sorting, BFS/DFS, binary search on answer, DP state transition, stack/queue simulation.

## Answer Review Table

| Mark | Meaning | Next Action |  
| Solved | correct under timer | revise after 7 days |  
| Guessed | answer came without method | solve two similar questions |  
| Wrong | concept or execution gap | write mistake rule and reattempt tomorrow |  
| Skipped | time or confidence issue | learn shortcut or move to lower difficulty |

## Execution Dashboard

| Metric | Target | Review Frequency |  
| DSA Questions | 10-14 per week | every Sunday |  
| CS Core Notes | 4 topics per week | twice a week |  
| Mock Tests | 1-2 per week | same day analysis |  
| Interview Practice | 5 answers per week | record and review |

## Weekly Review Ritual

- Pick the top 5 mistakes, rewrite the correct rule, and solve one similar question.
- Remove one weak resume line or prepare one stronger project answer.
- Decide the next week from evidence, not mood.

## Final Self Audit

| Score | Meaning | Action |  
| 0-40 | reading stage | finish basics and easy questions |  
| 41-70 | practice stage | increase timed mocks and mistake review |  
| 71-85 | interview-ready | polish project, HR, and weak topics |  
| 86-100 | selection-ready | maintain revision and avoid overloading new material |

## Source Note

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